Optimal Sequence Planning for Robotic Sorting of Recyclables From Source-Segregated Municipal Solid Waste

Article *in* Journal of Computing and Information Science in Engineering · June 2020

DOI: 10.1115/1.4047485

Sathish Paulraj Gundupalli
Vellore Institute of Technology University

Rishabh Shukla
University of Southern California

Rohit Gupta
Indian Institute of Technology Patna

Subrata Hait
Indian Institute of Technology Patna

Optimal Sequence Planning for Robotic Sorting of Recyclables from Source-segregated Municipal Solid Waste

Sathish Paulraj Gundupalli [a], Rishabh Shukla[b], Rohit Gupta[c],
Subrata Hait[d], Atul Thakur [e,*]

[a] *Department of Design and Automation, Vellore Institute of Technology, India*

[b,c,e]*Department of Mechanical Engineering,* [d]*Department of Civil and Environmental Engineering, Indian Institute of Technology Patna, India*

Abstract

Sorting of recyclables from source-segregated municipal solid waste (MSW) stream is an essential step in the recycling chain in a material recovery facility (MRF) for waste management. Manual sorting of recyclables in an MRF is a highly hazardous operation for human health as well as time-consuming. Application of robotics for automated waste sorting can alleviate these problems to a large extent. The total sorting time depends upon the pick-and-place (PAP) sequence used in a robotic sorting system. In this context, the generation of optimal PAP sequence plan is a key challenge considering that it cannot be solved by an exhaustive search due to the combinatorial explosion of the search-space. This paper reports an approach for generating optimal PAP sequence plan for robotic sorting of recyclables from source-segregated MSW stream in a system equipped with thermal-imaging technique. The PAP sequence generation is formulated as an optimization problem wherein the objective is to minimize the total sorting time. The formulated problem has been solved using a genetic algorithm (GA) based approach. Numerical simulations, as well as physical experiments using a 6-DOF articulated manipulator, has been performed to test and validate the developed optimal sequence generation algorithm. Results revealed an improvement of up to 4.28% speedup in total sorting time over that of randomly generated sequences. It is envisaged that the developed approach can substantially improve the sorting performance in an MRF.

Keywords: *municipal solid waste, source-segregation, multi-material sorting, robotic manipulation, pick-and-place sequence planning, genetic algorithm.*

[*] *Corresponding author*

[*] *Tel:* +91-612-3028158; *E-mail* addresses: [a]sathish.gp@vit.ac.in, [b]rishabh.me15@iitp.ac.in, [c]rohit.me15@iitp.ac.in, [d]shait@iitp.ac.in, [e,*]athakur@iitp.ac.in

1 Introduction

There is a rapid rise in MSW generation globally [1,2], especially in the lower to lower-middle income group countries, owing to the widespread urbanization and industrialization [1,3–5] (see Figure 1(a)). MSW is usually a rich source of various useful dry recyclable materials like metal, paper, plastic, and glass (see Figure 1(b)). The prospect of recycling has become more relevant from the perspective of MSW management. Recycling of useful materials from municipal solid waste (MSW) is economically promising as well as can help in conserving natural resources.

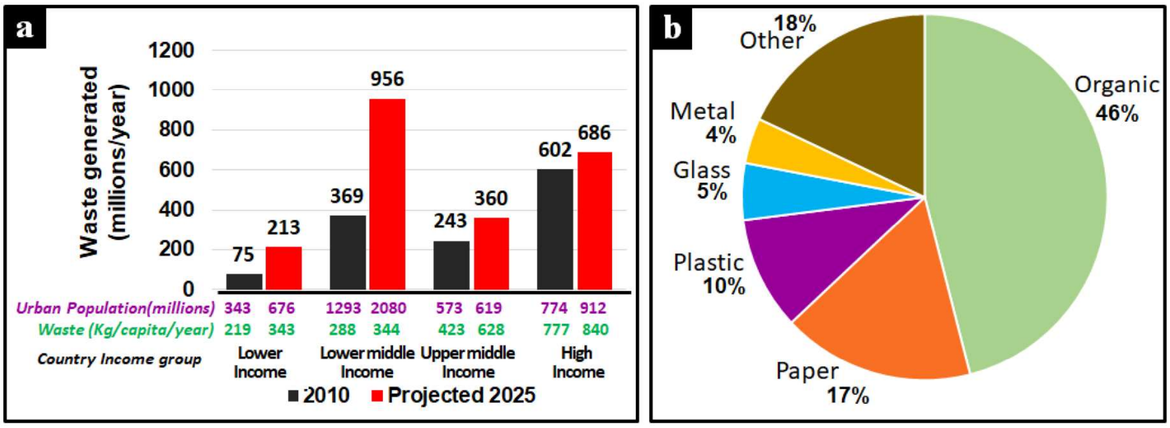


Figure 1: (a) Urban waste generation by income level, (b) Global MSW composition

Waste sorting is an essential step in the recycling chain in a material recovery facility (MRF). The workers involved in manual sorting of recyclables in an MRF, especially in the developing countries, are confronted with an unhygienic working environment, wherein they come in direct contact with dust, smell, noise, and hazardous substance. Some attempts to automate the sorting facilities have been made [6–9], but minimal research has been performed in the automated sorting of recyclables from MSW using a robotic system. In a robotic sorting system, the manipulation time depends upon the order in which the objects are picked and placed (PAP). The optimal PAP sequence planning of the objects is an instance of traveling salesman problem (TSP) and cannot be solved by an exhaustive search due to the combinatorial explosion of the search-space. Unlike printed circuit board (PCB) disassembly [10], the number of objects and the recyclable categories in the case of MSW sorting is very large which makes the problem of optimal PAP sequence planning cumbersome. Another challenge in the case of MSW sorting is the need for high computational performance of the optimal PAP sequence planning for the faster operation. In this context, this paper reports an approach for generating optimal PAP sequence plan for robotic sorting of recyclables from source-segregated MSW stream in a system equipped with thermal-imaging technique.

The developed system comprises of two modules namely: (i) inspection zone and (ii) manipulation zone. In the inspection zone, thermal imaging-based classification of the recyclable objects present in the MSW stream is performed [11]. We employ a k-nearest neighbor (KNN) based classification scheme while using a feature vector comprising of the mean and standard deviation of the thermogram representing the recyclable objects and also determine the respective poses. In the manipulation zone, the already classified objects carried by the conveyor are picked-and-placed by the robot to their respective bins. We developed an approach based on the genetic algorithms (GA) to generate the optimal PAP sequence to minimize the total sorting time. The paper also presents the details of the fitness function based on total sorting time computed using kinematic simulation.

2 Literature review

The problem of automated sorting has a long history in the industry, with the first sorting system for tomatoes dating back to the '70s [12]. Since then it has received significant attention from the robotics researchers, and a number of automated sorting techniques combining computer vision and robotic manipulation have been proposed.

A vast body of literature exists in the area of picking-and-placing of components in PCB assembly applications wherein the objective is to find the optimal sequence of components to be picked and placed for minimizing the total assembly time [10,13–15]. Elsayed et al. reported a technique for selective disassembly sequence planning for end-of-life electronic products problem [10]. It has been reported that the problem of disassembly sequence planning is an instance of a traveling salesman problem and is nondeterministic polynomial (NP) hard combinatorial optimization problem [16]. In general, the computational cost of searching for the optimal disassembly sequence plan increase exponentially with an increase in the number of components in the target PCB.

Consequently, metaheuristic methods are often used to find out near-optimal solutions [17,18]. The problem of PCB disassembly is conceptually close to the problem of sorting of recyclable materials from MSW [10]. However, the approach reported in [10] cannot be used directly as the fitness function uses time and its computed using the Euclidean distance which gives an over-conservative estimate of the distances between the components and their respective bins. Whereas we determined it using the simulation which gives much better estimates of the fitness function.

Optimization based on genetic algorithm (GA) is used for its capability to evolve toward an optimal solution without exhaustively processing all the alternatives. Lazzerini et al. [19], in their GA-based method for assembly planning, used chromosomal encoding comprising of disassembly sequence of components, directions of disassembly operations, and used gripper. Lukka et al. [20] employed machine learning based algorithms for the sorting of valuable recyclable materials from construction and demolition waste (CND) wherein the objective of sequence planning was to maximize the monetary value of the recovered recyclables. A combination of the vision sensor, 3D scanning, spectrum, and other sensors was used, and human annotations were used to improve the learning system continuously [24].

Ball and Magazine [21] presented a chip insertion problem to model the PCB component sequencing problem for a PAP machine. William and Ping [22] modeled the PAP sequencing problem as a TSP with a variation in the cost function. Unlike TSP, the target in the PAP sequencing problem is not to minimize the distance between individual components but to optimize the total sum of all the distances traveled. The process involves picking components from the feeder and placing it to the robot workspace. Therefore, the cost function has to be inclusive of the feeder position as well as component positions.

Chang et al. [23] designed a GA-based algorithm with external self-evolving multiple archives (ESMA) for PAP sequencing problem in PCB assembly. They also designed a clustering strategy, switchable mutation, and elitist propagation methods for GA execution. Christensen et al. [24] presented a detailed review of multi-dimensional bin packing problem. An object placement planner taking into account the convexity, contact and the stability test has been presented by Harada et al. [25]. Pellicciari et al. [26] presented a technique for generating energy-optimal trajectories for performing pick-and-place operations.

Most of the existing literature reports the techniques to solve the optimal pick-and-place problem relevant to the warehouse bin packing and PCB assembly/disassembly. However, research in the area of robotic segregation of MSW is very scant and mostly not available in the public domain. For example, Zenrobotics® started sorting municipal solid waste using RGBD camera augmented with a hyperspectral imager for classification [20,27]. A patent on thermal imaging based refuse separator for recyclables has been reported, but the system detail and the algorithms used have not been reported [28]. The research on sequence planning for disassembly is the closest to the presented approach. This paper presents a GA-based optimal sequence generation technique for sorting of recyclables from source-segregated MSW into four broad categories, namely iron, plastic, aluminum, and wood as classified using thermal imaging technique by Gundupalli et al. [11].

3 System overview and problem statement

3.1 System overview

A schematic representation of the thermal imaging-based sorting system is illustrated in Figure 2. To solve the problem of robotic sorting of recyclables from source-segregated MSW, we follow a two-step approach namely, (i) thermal imaging-based classification (performed in the inspection zone), and (ii) robotic sorting (performed in the manipulation zone).

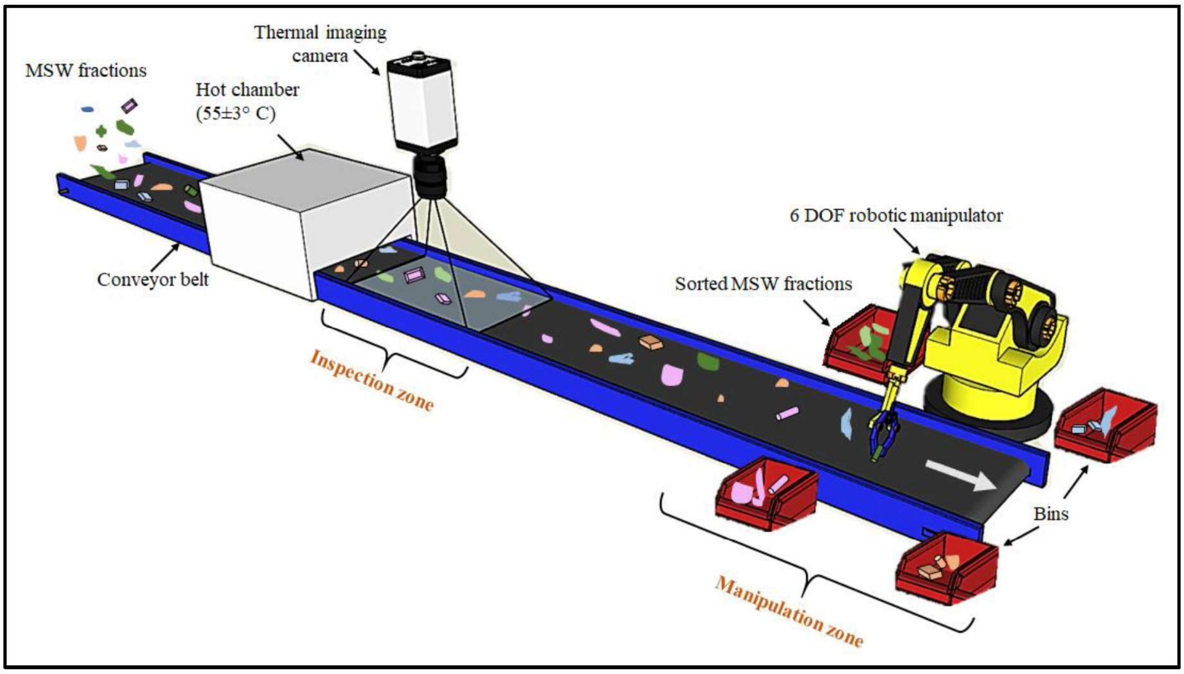


Figure 2: Schematic representation of the thermal imaging-based sorting system

In the inspection zone, the identification of recyclables from an MSW stream is performed using the thermal imaging technique (see Figure 2). After this, the identified recyclables are transferred to the manipulation zone. An overview of the algorithm in an inspection zone is illustrated in Figure 3. First, the thermograms are captured and normalized according to the workspace dimensions (see Figure 3). Second, we need to define the features of thermogram. According to the principle of thermal imaging, we know that objects with high emissivity will radiate more compared to less radiation emitted from low emissivity objects (see Figure 3). For example, a glass plate with an emissivity of 0.92 [29] will radiate more compared to an aluminum plate with an emissivity of 0.04 and will thus have a comparatively brighter thermal image [30–32]. To perform this, the feature extraction is performed using Otsu's thresholding method [11,33] to determine the image segments representing the recyclable objects (see Figure 3). Third, the image segments are processed to identify features like mean and standard deviation of intensities are presented to the classifier model to determine the type of recyclable material that the image segment represents (classified objects like iron, aluminum, plastic, and wood). Fourth, the boundary contours of the image segments followed by the poses of the classified recyclable object are computed, which is subsequently used for sorting (see Figure 3). The thermograms of MSW and the corresponding histogram plot of the identified dry recyclable object in the scene with the bounding box can be seen in Figure 3. The details and operation of the developed approach for the inspection zone are discussed by Gundupalli et al. [11].

In the manipulation zone, the identified MSW fractions are fed to the robot workspace via the conveyor belt (see Figure 2). Subsequently, the robotic system performs the PAP operation of the recyclables into their respective bins. In the manipulation zone, the already classified objects carried by the conveyor are picked and placed by the robot to their respective bins (see Figure 3). To minimize the total sorting time, we transferred the object poses and fitness function to the genetic algorithm. Later, we obtained the optimal sequence of recyclable objects to be picked that can minimize the overall sorting time. The sorting time for a given object is determined as a sum of the time taken for the manipulator to reach to the grasping location, the grasping time, and the time taken for the manipulator to reach the placing point, i.e., near to the respective bin location, and the placing time. Followed by the trajectory plan is determined and sent to the 6 DOF robotic manipulator for PAP operation (see Figure 3). The main objective is to determine the optimal sequence that minimizes the total sorting time. A formal description of the aforementioned problem is presented in section 3.2.

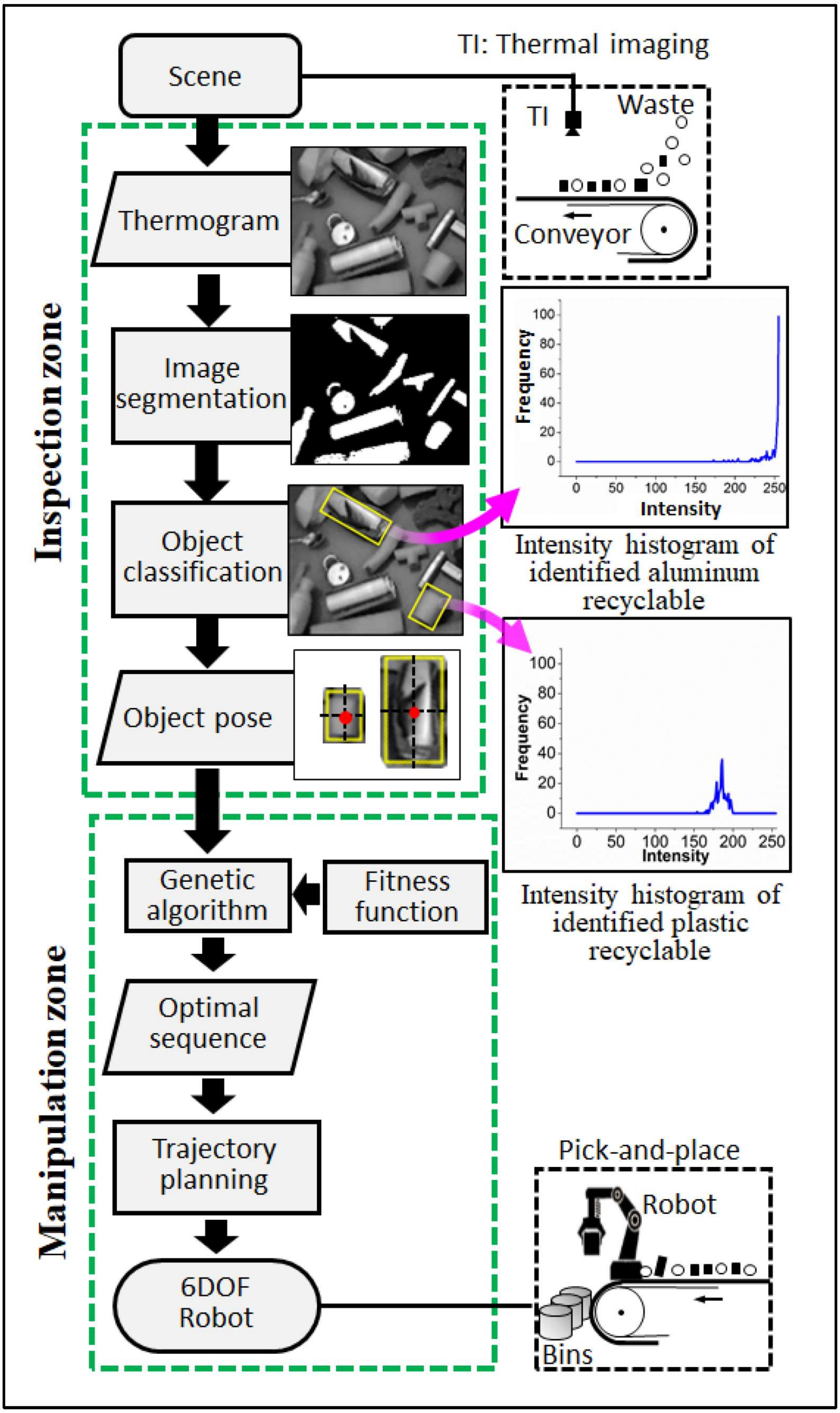


Figure 3: Overview of the developed approach for robotic sorting of recyclables from source-segregated MSW

3.2 Problem statement

Let us suppose that there are n objects present in the robot's workspace namely $O_1, O_2, \dots O_n$ (see Figure 4). Each object O_i can belong to one of the $m < n$ material classes $T_1, T_2, \dots T_m$ which is already known using the approach presented in [11]. Let the material class that the object O_i belong to be given by $C_{O_i} = cat(O_i)$. The objects O_i are located at $pose(O_i) = {}^{\Lambda}x_{O_i} = {}^{\Lambda}_{\Omega}T\, {}^{\Omega}x_{O_i}$ where ${}^{\Lambda}_{\Omega}T$ is the transformation matrix between the global frame Λ and the local frame Ω (see Figure 4). We assume that ${}^{\Lambda}x_{O_i}$ is located inside the robot workspace. We assume that corresponding to each material type T_j there is a unique bin B_{T_j} located at $pos\left(B_{T_j}\right) = {}^{\Lambda}x_{T_j}$ which lies inside the robot workspace.

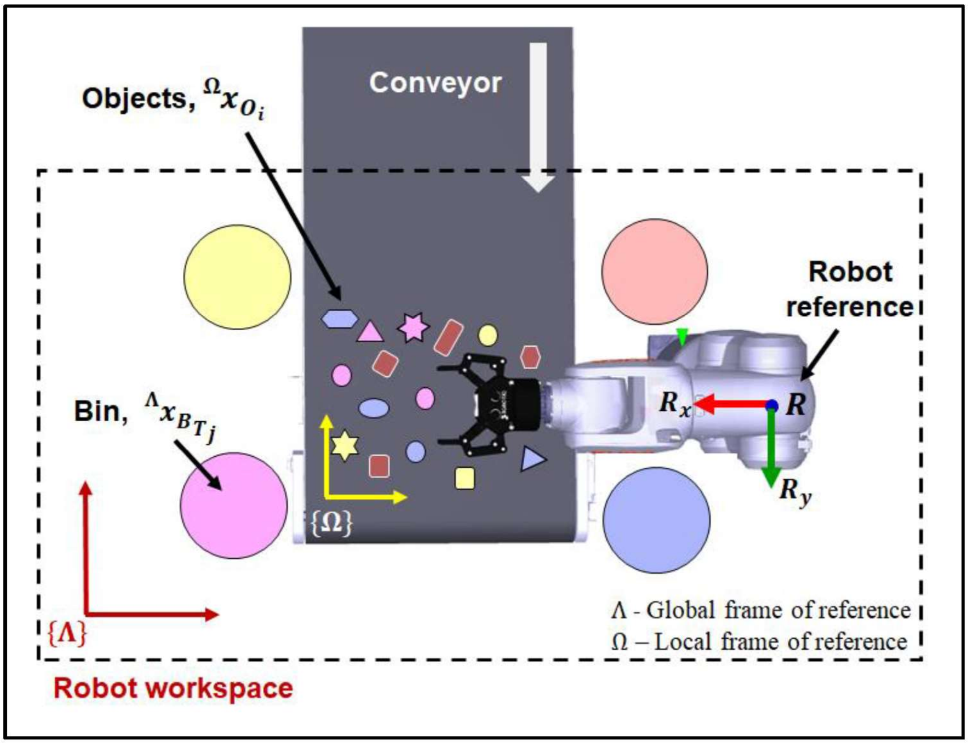


Figure 4: Schematic representation of the problem statement

Let the set $O = \{O_1, \dots O_n\}$ represents a given scene. We define a sequence $S = \{S_1, S_2, \dots S_n\} = perm(O)$ which is an arbitrary permutation over O and may be used for sorting the objects $S_k \in O$ into their respective bins $B_{cat(S_k)}$. We now define the total sorting time (Eq. 1) for a given sequence $S = \{S_1, S_2, \dots S_n\}$ as,

$$t(\mathrm{s}) = \mathrm{Time}\left(h, x_{S_1}\right) + nt_{pick} + \mathrm{Time}\left(x_{S_1}, pos(B_{cat(S_1)})\right) + nt_{place} + \sum_{i=2}^{n} \left(\mathrm{Time}\left(pos(B_{cat(S_{i-1})}), x_{S_i}\right) + \mathrm{Time}\left(x_{S_i}, pos(B_{cat(S_i)})\right)\right) \quad (1)$$

where h is the robots home position, and t_{pick} and t_{place} are the times taken for grasping and placing, respectively, and $\mathrm{Time}\left(h, x_{S_1}\right)$ is the time taken for robot to move from h to object location x_{S_1}.

The objective is to determine the optimal sequence $S^* = \underset{S}{\mathrm{argmin}}\, t(S)$.

There are four tasks to be performed for moving an object to bin:

- $Move\ (pos(B_{cat(O_{i-1})}),\ x_{O_i})$: moving end-effector from the bin location $pos(B_{cat(O_{i-1})})$ to the object with pose x_{O_i}, where $pos(B_{cat(O_{i-1})})$ is the bin location where the previous object was placed. The time taken to move between these two points is determined by the simulator.
- $Grasp\ (H_1, x_{O_i})$: Grasping object x_{O_i} with H_1 configuration of the end effector.
- $Move\ (x_{O_i}, pos(B_{cat(O_i)})\ )$: moving the object with $pose(O_i)$ to $pos(B_{cat(O_i)})$, where $B_{cat(O_i)}$ is the bin corresponding to the material category of object O_i.
- $Place\ (H_2, O_i, B_{cat(O_i)})$: Placing the object O_i to its corresponding bin $B_{cat(O_i)}$ with the end effector configuration H_2.

We observed that the major time is spent on moving the end effector to the object and then to the bin and which is varying based on the bin location and the object pose. The aforementioned approach for determining the sorting time has been described using an example in which sorting time corresponding to a given sequence S is determined (see Figure 5).

Consider a scene comprising of seven objects $O = \{O_1, O_2, O_3, O_4, O_5, O_6, O_7\}$ for which the sorting time corresponding to an arbitrary PAP sequence represented as $S = \{O_1, O_3, O_5, O_6, O_4, O_7, O_2\}$ is to be determined. Let us suppose that the material category of O_2 is T_1; material category of O_1 and O_4 is T_2; material category of O_3 and O_6 is T_3; and the material category of O_5 and O_7 is T_4.

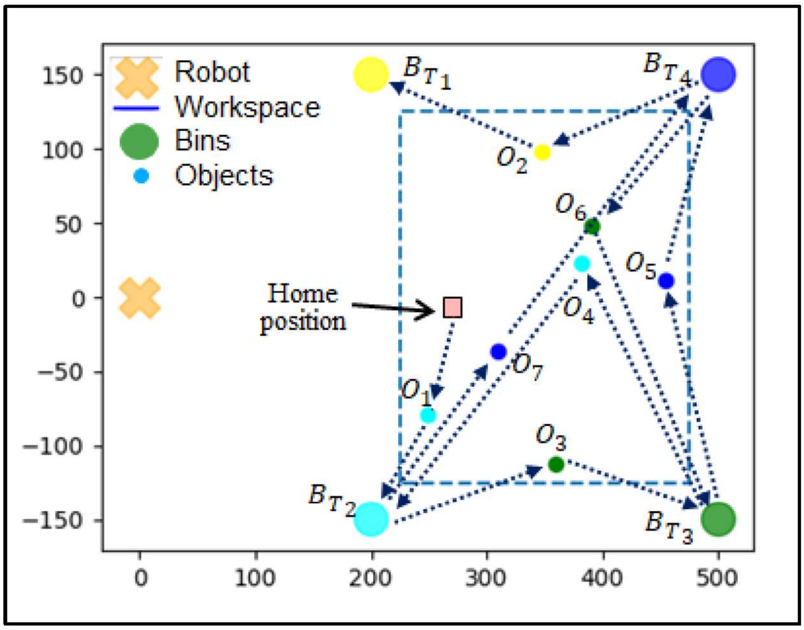


Figure 5: Illustrative example for describing the computation scheme to determine sorting time for a given PAP sequence $S = \{O_1, O_3, O_5, O_6, O_4, O_7, O_2\}$

Thus, for the given PAP sequence S, the robot should move from the robot home position to O_1 (see Figure 5). After that, it should $Grasp$ the object O_1 and then $Move$ from O_1 to the bin location B_{T_2}. Subsequently, the robot should $Place$ O_1 into the bin B_{T_2}. Similarly, rest of the PAP sequence i.e. $O_3 \rightarrow B_{T_3} \rightarrow O_5 \rightarrow B_{T_4} \rightarrow O_6 \rightarrow B_{T_3} \rightarrow O_4 \rightarrow B_{T_2} \rightarrow O_7 \rightarrow B_{T_4} \rightarrow O_2 \rightarrow B_{T_1}$ should be executed (see Figure 5). To compute the time required for executing the PAP sequence has been determined using RoboDK™ simulator with a 6-DOF articulated manipulator in this paper. The objective is to determine the PAP sequence for which the total time required to execute the sequence is minimized.

In the following sections, the approach for generating the optimal sequence has been presented. First, the details of dynamics simulation-based computation of robot trajectories for determining the sorting time required for performing PAP operation has been presented. After this, the details of GA-based optimal sequence planning have been described. Finally, the results of numerical simulations and experimental simulations for validating the developed approach has been presented.

4 Optimal sequence planning for sorting

4.1 Dynamics simulation-based computation of pick-and-place times

To determine the fitness function which is used in optimization we employed dynamics simulations carried out in RoboDK™ software. Simulation instead of physical experiments is particularly useful in the repetitive evaluation of fitness functions due to (1) being inexpensive and (2) causing less wear and tear to the physical robotic system. However, it must be ensured that the error between the values of fitness function determined by simulations and experiments is negligible.

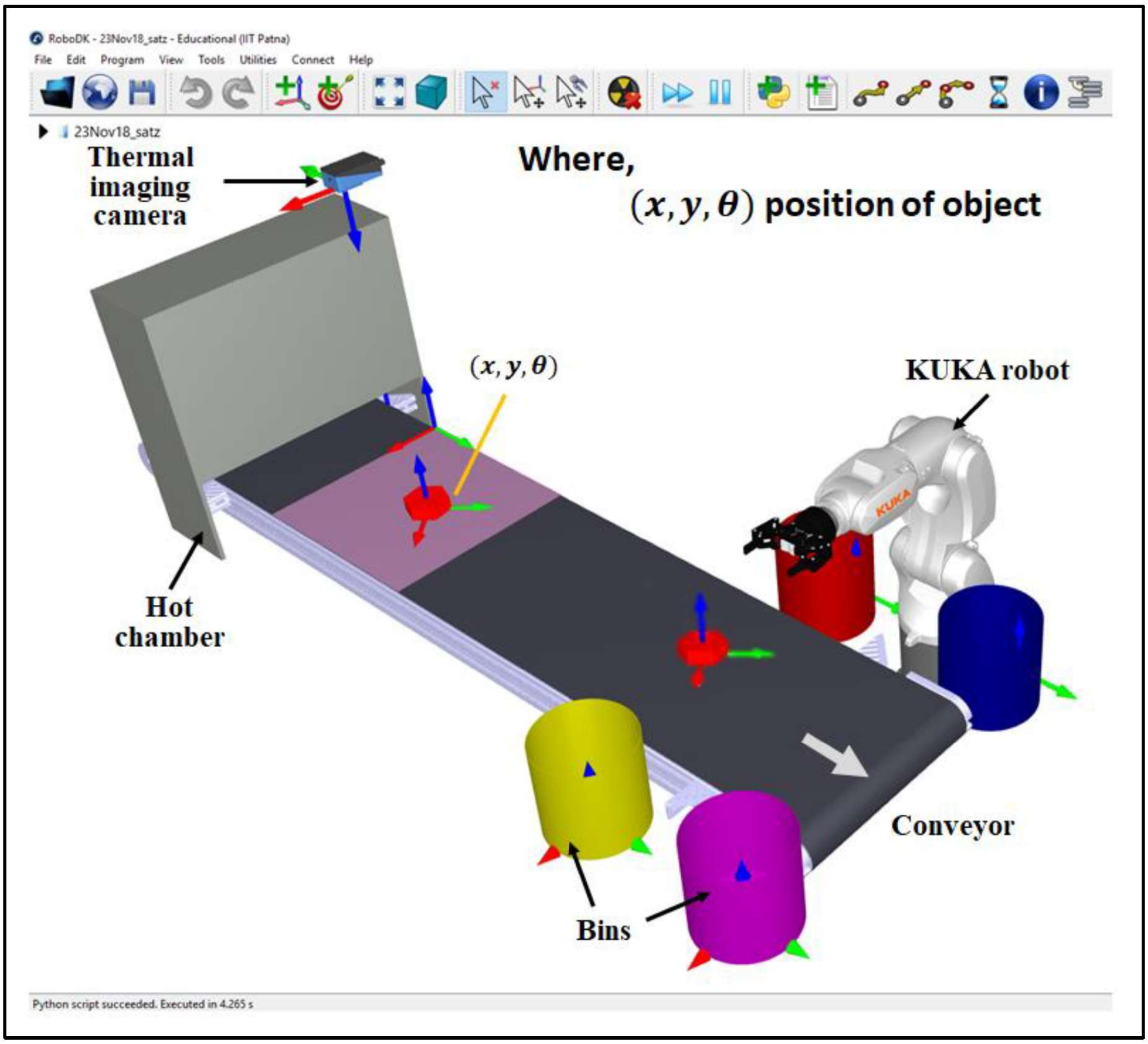
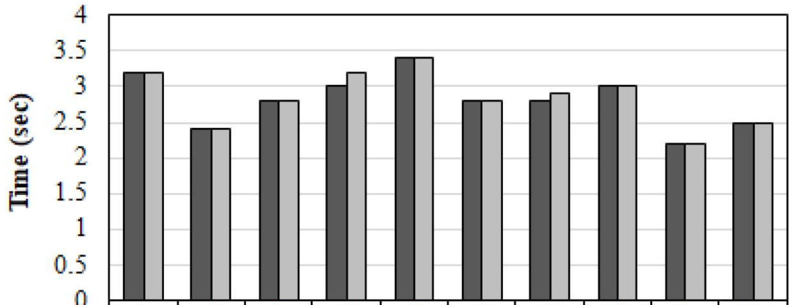


Figure 6: Complete setup in a simulation environment (RoboDK™ software) with various components.

Figure 6 shows the complete setup in a simulation environment (RoboDK™). The 3D models of components like conveyor and chamber are imported in the simulation work environment. We replicated the real work environment in the simulator to perform the experiments.

We initially generated a random scene of objects with their pose and material types that represent the real environment in the inspection zone (see Figure 6). The generated objects are transferred to robot workspace via the simulated conveyor. Subsequently, the objects are sorted by the simulated robot into their respective bins as per a given PAP sequence. The simulator can check for possible collisions between the robot and other elements in the work environment. The simulator provides a simple script-based programming interface.



Case	1	2	3	4	5	6	7	8	9	10
Time taken by robot	3.2	2.4	2.8	3	3.4	2.8	2.8	3	2.2	2.5
Time calculated in simulation	3.2	2.4	2.8	3.2	3.4	2.8	2.9	3	2.2	2.5

Case

Figure 7: Comparison between pick-and-place times predicted by the simulation and determined from the physical robotic system.

For a given PAP sequence, the sorting time is computed using RobotDK™ which is used as the fitness value in GA-based optimization. To determine the error between the total sorting times predicted by the simulator with respect to that obtained on the physical robotic system we conducted several experiments for PAP operation. Time taken to pick target objects located at ten random poses followed by placing them into the bin was recorded both for the simulation and the physical setup. Figure 7 illustrates the comparison of PAP times taken in the simulation and the physical experiments. We observed an average error of around 1% between the PAP time predicted by the simulator and that obtained from the physical experiments. Due to the observed negligible error, simulation-based fitness estimation has been used for optimization.

4.2 GA-based optimal sequence planning

4.2.1 Chromosomal encoding and fitness computation

Chromosomal encoding

Chromosomal encoding is a way of representing solutions in GA. Every object present in the given scene is assigned a unique index. We represent a solution/chromosome as a permutation of the aforementioned indices representing the objects present in the scene. A database comprising of the information about the unique index, $pose$, and category of each object is formed using the techniques developed in [11]. As the name signifies the object category refers to the bin in which the object has to be placed. Figure 8 illustrates the chromosomal representation for a PAP sequence of 10 objects.

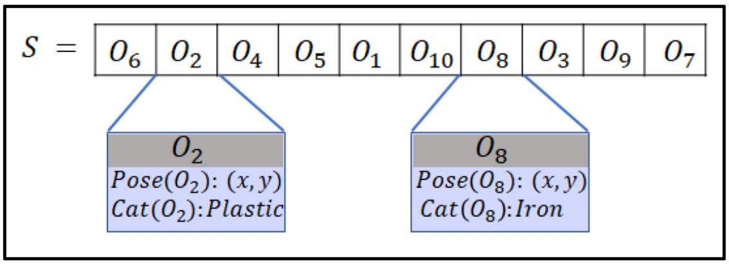


Figure 8: Representation of an individual chromosome used for GA evaluation

Fitness function

The fitness function returns the time taken for the PAP operation of an object sequence represented by a chromosome. The fitness function in the study is the same as the sorting time function (see Eq.1). The developed simulation-based estimation of the fitness function instead of Euclidean distances is more accurate as the discontinuities in the joint motions and accelerations are taken into account while computing. The offline simulation-based estimation of the fitness function provides nearly 1% accuracy as described in section 4.1.

Although simulation gives an accurate estimate, but real-time computation of that is time-consuming. Therefore, we have precomputed all the possible cases and stored them into a database: $Database\,\{S_i, B_{Tj}\} = Time\,\{\,pose(S_i), pose(B_{Tj})\}$. The fitness function, i.e., $Fitness\,\{S\} = t\,\{S\}$, can then be determined using (Eq. 1), wherein the times of individual robot motions can be determined from the databasse.

We have ignored the time required for closing and opening of the gripper in the fitness calculation because it is a constant factor throughout and can be significantly improved by the use of vacuum grippers. The possible sources of error as mentioned earlier are the use of offline simulation instead of the real robot (since the robot are prone to wear and tear), and error introduced due to the discretization of the workspace.

4.2.2 Selection of GA operators

Population: A set of several chromosomes makes a population. The population size is kept ten times the size of an individual, which is an empirical rule for optimal results [34].

Selection operator: We have used a tournament selection operator. Tournament selection executes many "tournaments" among a few randomly selected individuals from the population. The individual with the best fitness in a tournament is selected for *crossover*. We selected one winner at the end of each tournament; hence we have to run several tournaments in order to choose the same number of offspring as the population. The number of chromosomes competing in each tournament is called tournament size, which is set to be 10 in this paper.

Crossover operator: We used partially matched crossover (PMX) operator in which the two chromosomes are aligned and randomly two crossover points are selected. The crossover probability was chosen as 0.9.

Mutation operator: For the reported experiment, we have used a mutation probability of 0.0005 to facilitate diversity in the population. Table 1 summarizes the parameter selection for the GA-based optimization.

Table 1: A summary of GA parameters used

Genes	Individual objects
Chromosome size	10 to 100
Population size	10 × Chromosome size
Fitness function	Eq. 1
Selection Operator	Tournament selection with size 10
Mutation Operator	Shuffling indexes with a probability of 0.0005
Crossover Operator	Partially matched with crossover probability 0.9

5 Validation

5.1 Numerical simulations

We generated one-hundred scenes each for 10, 15, 20, …, 100 objects wherein the objects were placed randomly by varying the randomization seed values. After this, optimal sequence corresponding to each scene was generated using the developed GA-based approach. The bins are represented by yellow, blue, pink, and red colored square boxes. The recyclable objects present in the environment are represented with different colors i.e., yellow, blue, pink, and red denoting material types. The bins B_{T_j} are placed in each corner of the robot workspace.

We performed GA-based optimization for each of the aforementioned 1900 cases. For brevity, we depict the evolution of fitness for the case with 50 objects and their representative seed values $S_d = 1, 2,$ and 3 (see Figure 9). For $S_d = 1$, we obtained the manipulation time of 53 sec for a random sequence 'R_s' and the manipulation time of 50.24 sec for the corresponding optimized solution 'Op_s' (see Figure 9). Thus, we observed an improvement of 5.22% in the sorting time.

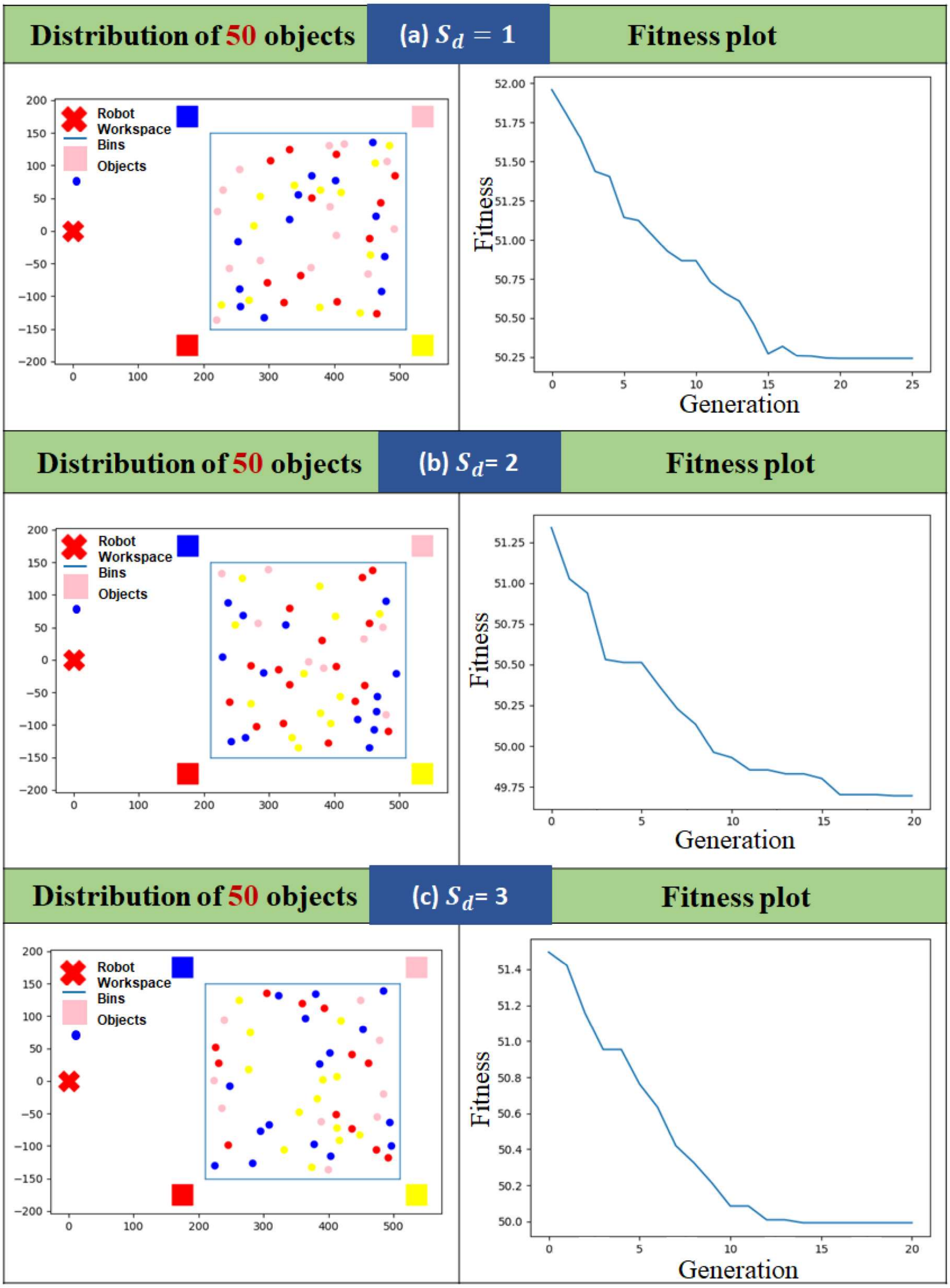


Figure 9: A representative case of evolution of fitness for a scene with 50 objects for (a) $S_d = 1$, (b) $S_d = 2$, and (c) $S_d = 3$

Similarly, for $S_d = 2$, R_s = 52.37 sec, Op_s = 49.69 sec, with 5.12% improvement in the sorting time and $S_d = 3$, R_s = 52.69 sec, Op_s = 49.99 sec, with 5.13% improvement in the sorting time were observed. When the GA-based optimization was run for all the 100 seed values the scene containing 50 objects, we obtained an average improvement of 4.34% in the sorting time over random sequence.

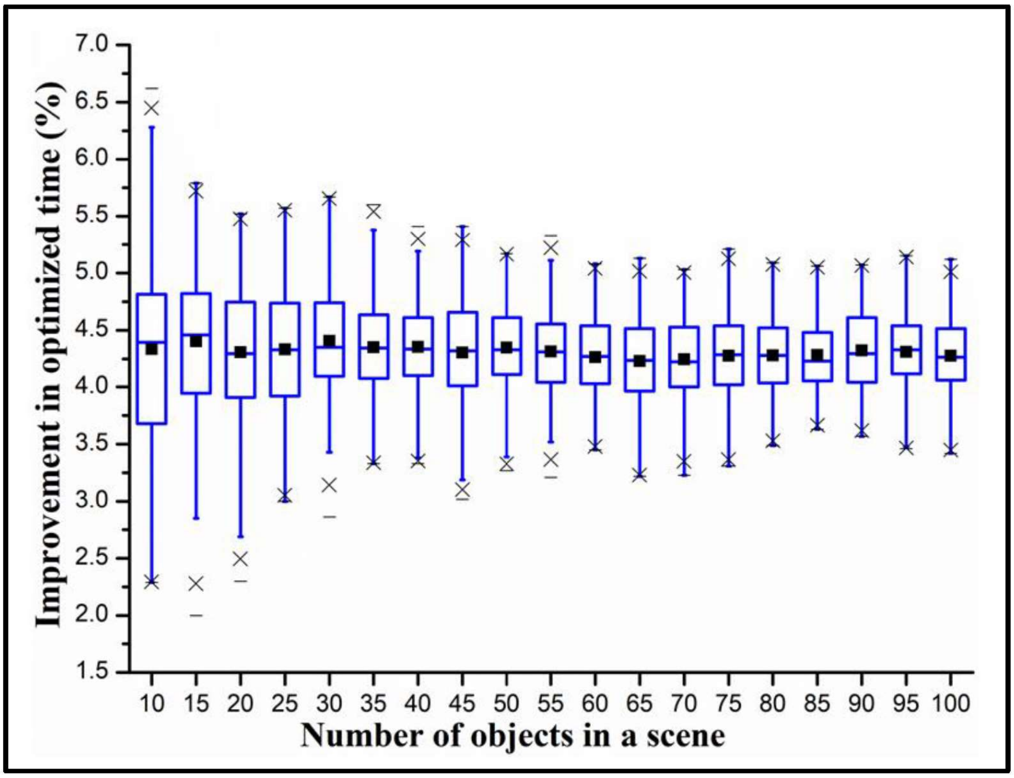
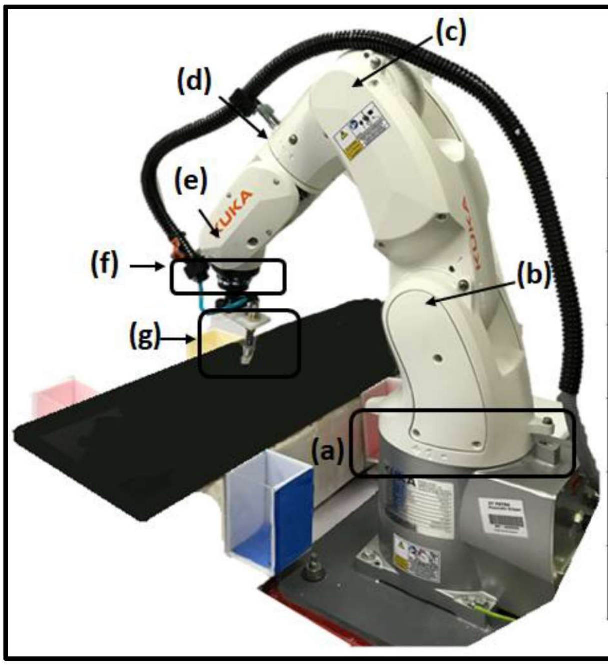


Figure 10: Variation of percentage improvement in optimized time for a different number of objects

We repeated the aforementioned optimizations for other cases, i.e., scenes containing 10, 15, 20,…, 100 objects and the corresponding percentage improvements in the total sorting times over random sequences have been illustrated in Figure 10. It is evident that the improvement in the total sorting time ranges between 2.29% to 6.62% for the reported experiments. The average improvement of 4.28% was obtained for the reported experiments.

5.2 Experimental validation

The study uses a 6 DOF robot (KUKA KR3 Agilus) manipulator that has 3 kg payload capacity and 500 mm reach to test the optimal sequence in the physical environment (see Figure 11). The robot manipulator includes nine parts: (a) a base joint, (b) a shoulder joint, (c) an elbow joint, (d) an upper arm joint, (e) a wrist joint, (f) a wrist rotation joint, and (g) a pneumatic gripper. A servo-motor is used for actuating each joint of the manipulator. The robot is equipped with a pneumatic gripper. To grasp the objects of varying shapes and orientations, a vacuum gripper has been employed.



D-H parameter of robot

	$\alpha°$	a (mm)	d (mm)	$\theta°$	$\theta°_{\min}$	$\theta°_{\max}$
1	0	0	345	θ_1	−170	170
2	−90	20	0	θ_2	−170	50
3	0	260	0	$\theta_3 - 90$	−110	155
4	−90	20	260	θ_4	−175	175
5	90	0	0	θ_5	−120	120
6	−90	0	75	$\theta_6 + 180$	−350	350

Figure 11: Hardware of the 6-DOF robot manipulator used in the object PAP experiments and its D-H parameters. Robot manipulator includes nine parts: (a) a base joint, (b) a shoulder joint, (c) an elbow joint, (d) upper arm joint, (e) a wrist joint, (f) a wrist rotation joint, and (g) a vacuum gripper.

Figure 12 illustrates the experimental setup of the robotic manipulator-based PAP operation using the source-segregated MSW. The robotic sorting of recyclable objects from the source-segregated MSW stream to their respective bins has been demonstrated. For the given arbitrary scenes containing a sample of 10 recycle objects weighing 1 kg altogether, the robotic sorting is performed using the optimal sequence determined from the developed approach (see video in *Appendix Ext:1*). We observed an improvement of 4.3% in PAP time. The robot PAP speed used was 2770 picks/hr and an average throughput of up to 280 kg/hr.

To demonstrate the application of the developed technique on larger number of components, we used surrogate objects of smaller size. This is because the workspace of the robot for recyclable objects is small. Figure 13 illustrates the experimental setup of the robotic manipulator-based PAP operation using surrogate objects. The robotic sorting of surrogate objects to their respective bins has been demonstrated.

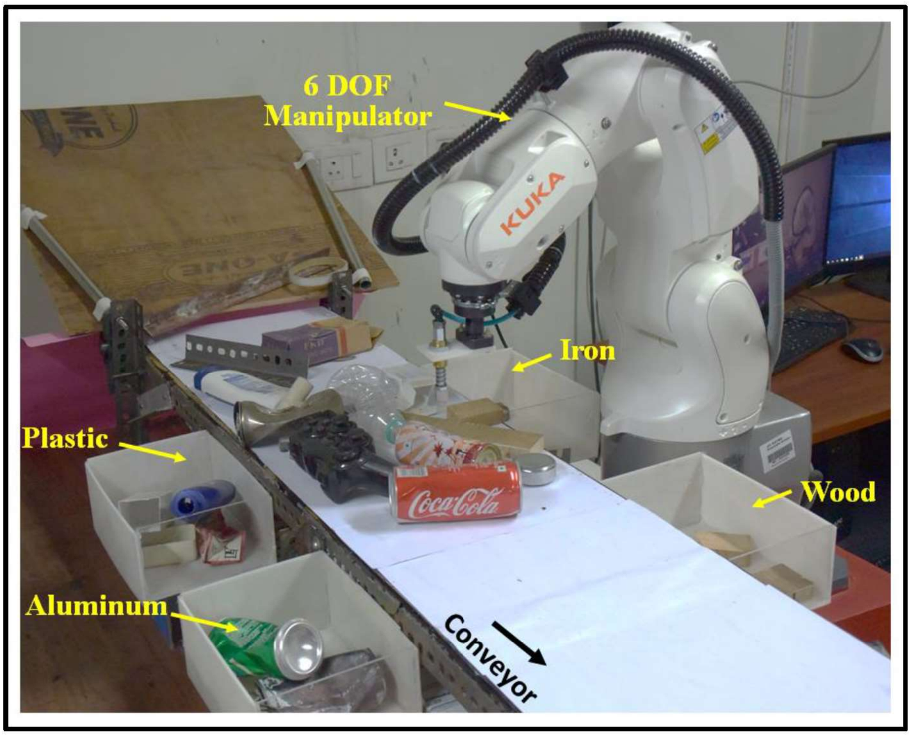


Figure 12: Robotic manipulator-based PAP experimental setup using source-segregated MSW.

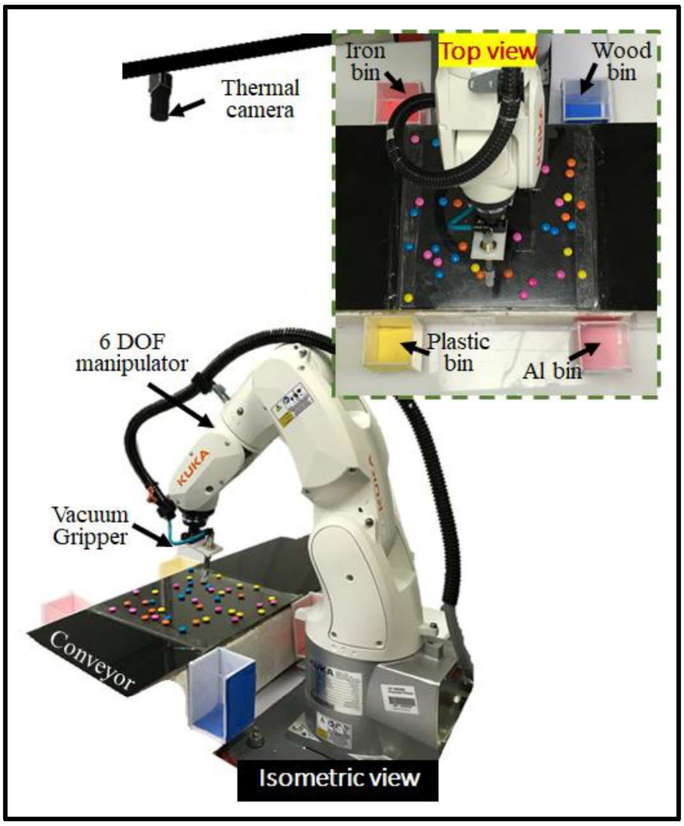
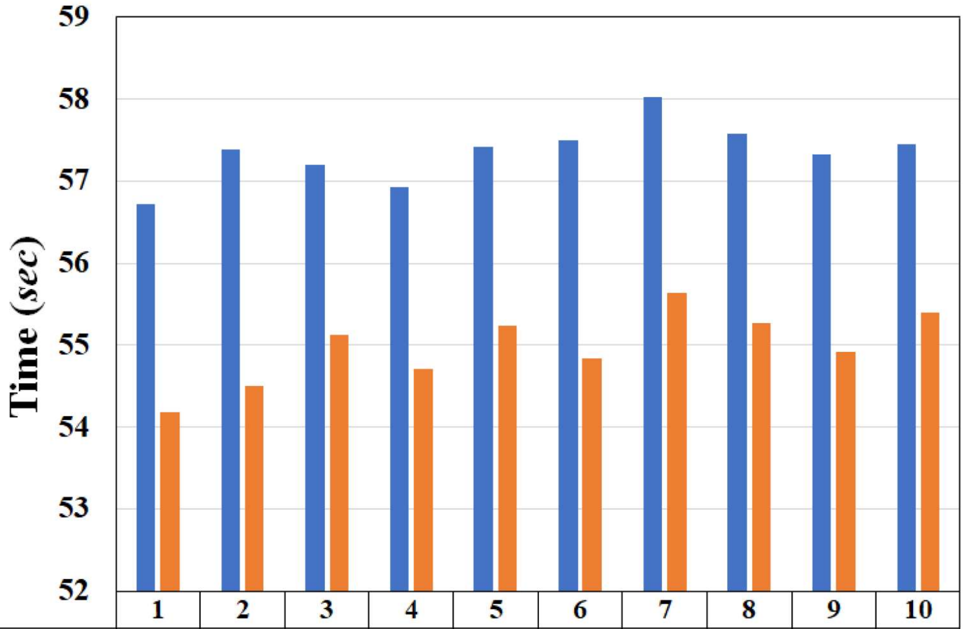


Figure 13: Robotic manipulator-based PAP experimental setup using 50 surrogate objects.

The surrogate objects present in the workspace are represented using different colors denoting various material types as follows: Red → iron, Yellow → plastic, Pink → aluminum, Blue → wood (see Figure 13). RoboDK™ trajectory planning module was used for the trajectory generation and control of the robot.

Figure 14 illustrates the comparison of the PAP times taken by the physical robot for sorting objects in 10 arbitrary scenes containing 50 objects with random sequences and the corresponding optimized sequences. We observed an average improvement of 4.13% (see Figure 14) using the optimized sequences obtained using the developed approach (see video in *Appendix Ext:2*).



	1	2	3	4	5	6	7	8	9	10
Randomized time	56.71	57.39	57.19	56.93	57.42	57.5	58.02	57.57	57.33	57.45
Optimized time	54.19	54.5	55.12	54.71	55.24	54.84	55.63	55.27	54.92	55.4
Improvement (%)	4.44	5.03	3.61	3.89	3.79	4.62	4.11	3.99	4.20	3.56

Figure 14: Comparison between PAP times predicted by the simulator and the physical robot system for 10 different representative scenes with 50 objects of 4 different materials.

6 Conclusions

This paper presents a GA-based optimal sequence planning algorithm for the robotic PAP sorting of recyclables from source-segregated MSW stream. A fitness function based on PAP times computed using dynamic simulations has been formulated and presented in this paper. Numerical simulation experiments for various scenes comprising of 10 to 100 objects were performed to validate the developed approach. An average improvement of 4.28% speedup in the total sorting time was obtained using the developed approach over that of a randomly generated sequence. We also performed validation experiments on a physical robotic setup with scenes comprising of 50 objects and observed an average speedup of up to 4.13% in the total sorting time which is in close agreement with the numerical simulation results.

The developed approach can find applications for improving the sorting performance in an MRF. The system performance can be improved by employing multiple robots and adapting the sequence planning algorithm accordingly. Further, dynamic pickup from the moving conveyor may be implemented for an enhanced throughput.

References

[1] Hoornweg, D., and Bhada, P., 2012, “What a Waste. A Global Review of Solid Waste Management,” Urban Dev. Ser. Knowl. Pap. no. 15. World Bank, Washington, DC. © World Bank.

[2] Kawai, K., and Tasaki, T., 2016, “Revisiting Estimates of Municipal Solid Waste Generation per Capita and Their Reliability,” J. Mater. Cycles Waste Manag., **18**(1), pp. 1–13.

[3] Chiemchaisri, C., Juanga, J. P., and Visvanathan, C., 2007, “Municipal Solid Waste Management in Thailand and Disposal Emission Inventory,” Environ. Monit. Assess., **135**(1), pp. 13–20.

[4] Liu, C., and Wu, X., 2011, “Factors Influencing Municipal Solid Waste Generation in China: A Multiple Statistical Analysis Study,” Waste Manag. Res., **29**(4), pp. 371–378.

[5] Saeed, M. O., Hassan, M. N., and Mujeebu, M. A., 2009, “Assessment of Municipal Solid Waste Generation and Recyclable Materials Potential in Kuala Lumpur, Malaysia,” Waste Manag., **29**(7), pp. 2209–2213.

[6] Holloway, C. C., 1989, “Method for Separation, Recovery, and Recycling of Plastics from Municipal Solid Waste,” U.S. Pat., (4,844,351).

[7] Roman, W. C., 1992, “Solid Waste Sorting System,” U.S. Pat., (5,101,977).

[8] Gundupalli, S. P., Hait, S., and Thakur, A., 2017, “A Review on Automated Sorting of Source-Separated Municipal Solid Waste for Recycling,” Waste Manag., **60**, pp. 56–74.

[9] Gundupalli Paulraj, S., Hait, S., and Thakur, A., 2016, “Automated Municipal Solid Waste Sorting for Recycling Using a Mobile Manipulator,” *Volume 5A: 40th Mechanisms and Robotics Conference*, p. V05AT07A045.

[10] Elsayed, A., Kongar, E., Gupta, S. M., and Sobh, T., 2012, “A Robotic-Driven Disassembly Sequence Generator for End-of-Life Electronic Products,” J. Intell. Robot. Syst. Theory Appl., **68**(1), pp. 43–52.

[11] Gundupalli, S. P., Hait, S., and Thakur, A., 2017, “Multi-Material Classification of Dry Recyclables from Municipal Solid Waste Based on Thermal Imaging,” Waste Manag., **70**, pp. 13–21.

[12] Husome, R. G., Fleming, R. J., and Swanson, R. E., 1978, "Color Sorting System," U.S. Pat., (4,131,540).

[13] Zhu, G.-Y., and Zhang, W.-B., 2014, "An Improved Shuffled Frog-Leaping Algorithm to Optimize Component Pick-and-Place Sequencing Optimization Problem," Expert Syst. Appl., **41**(15), pp. 6818–6829.

[14] García-Nájera, A., Brizuela, C. A., and Martínez-Pérez, I. M., 2015, "An Efficient Genetic Algorithm for Setup Time Minimization in PCB Assembly," Int. J. Adv. Manuf. Technol., **77**(5–8), pp. 973–989.

[15] Hui, W., Dong, X., and Guanghong, D., 2008, "A Genetic Algorithm for Product Disassembly Sequence Planning," Neurocomputing, **71**(13–15), pp. 2720–2726.

[16] Li, Y., and Chen, Y., 2010, "A Genetic Algorithm for Job-Shop Scheduling," J. Softw., **5**(3), pp. 269–274.

[17] Kongar, E., and Gupta, S. M., 2002, "Genetic Algorithm for Disassembly Process Planning," *Environmentally Conscious Manufacturing II*, pp. 54–62.

[18] Lit, P. De, Latinne, P., Rekiek, B., and Delchambre, A., 2001, "Assembly Planning with an Ordering Genetic Algorithm," Int. J. Prod. Res., **39**(16), pp. 3623–3640.

[19] Lazzerini, B., Marcelloni, F., Dini, G., and Failli, F., 1999, "Assembly Planning Based on Genetic Algorithms," *18th International Conference of the North American Fuzzy Information Processing Society-NAFIPS (Cat. No. 99TH8397)*, pp. 482–486.

[20] Lukka, T. J., Tossavainen, T., Kujala, J. V, and Raiko, T., 2014, "ZenRobotics Recycler--Robotic Sorting Using Machine Learning," *Proceedings of the International Conference on Sensor-Based Sorting (SBS)*, pp. 1–8.

[21] Ball, M. O., and Magazine, M. J., 1988, "Sequencing of Insertions in Printed Circuit Board Assembly," Oper. Res., **36**(2), pp. 192–201.

[22] Ho, W., and Ji, P., 2009, "An Integrated Scheduling Problem of PCB Components on Sequential Pick-and-Place Machines: Mathematical Models and Heuristic Solutions," Expert Syst. Appl., **36**(3, Part 2), pp. 7002–7010.

[23] Chang, P. C., Huang, W. H., and Ting, C. J., 2012, "Developing a Varietal GA with ESMA Strategy for Solving the Pick and Place Problem in Printed Circuit Board

Assembly Line," J. Intell. Manuf., **23**(5), pp. 1589–1602.

[24] Christensen, H. I., Khan, A., Pokutta, S., and Tetali, P., 2016, "Multidimensional Bin Packing and Other Related Problems: A Survey," pp. 1–34 [Online]. Available: https://pdfs.semanticscholar.org/bbcf/4ca2524cd50fdb03b180aa5f98d2daa759ce.pdf.

[25] Harada, K., Tsuji, T., Nagata, K., Yamanobe, N., and Onda, H., 2014, "Validating an Object Placement Planner for Robotic Pick-and-Place Tasks," Rob. Auton. Syst., **62**(10), pp. 1463–1477.

[26] Pellicciari, M., Berselli, G., Leali, F., and Vergnano, A., 2013, "A Method for Reducing the Energy Consumption of Pick-and-Place Industrial Robots," Mechatronics, **23**(3), pp. 326–334.

[27] "Zenrobotics" [Online]. Available: https://zenrobotics.com/solutions/applications/#materials-recovery-facilities.

[28] Thomas, D. W., 1997, "Thermal Imaging Refuse Separator," U.S. Pat., (5,628,409).

[29] Holst, G. C., 2000, *Common Sense Approach to Thermal Imaging*, JCD Publishing and SPIE - The International Society for Optical Engineering, Washington.

[30] Incropera, F. P., DeWitt, D., Bergman, T. L., and Lavine, A. S., 2006, "Fundamentals of Heat and Mass Transfer 6th Edition, 2006."

[31] Maldague, X., 2001, *Theory and Practice of Infrared Technology for Nondestructive Testing*, John Wiley and Sons, New York, NY, USA.

[32] Modest, M., 2003, "Radiative Heat Transfer 2nd Edn (New York: Academic)."

[33] Otsu, N., 1979, "A Threshold Selection Method from Gray-Level Histograms," IEEE Trans. Syst. Man. Cybern., **9**(1), pp. 62–66.

[34] Goldberg, D. E., and Holland, J. H., 1988, "Genetic Algorithms and Machine Learning," Mach. Learn., **3**(2), pp. 95–99.

Appendix: Index to Multimedia Extension

Extension	Media type	Description
1	Video	This video shows PAP operation corresponding to optimal sequence generated for 10 objects using source separated MSW stream.
2	Video	This video shows PAP operation corresponding to optimal sequence generated for 50 objects using surrogate objects.